

MA 442 - Quiz

April 1

Name: _____ BUID: _____

Question 1. Use elementary row operations to compute the determinant of

$$B = \begin{bmatrix} 1 & 1 & 0 & 2 \\ 2 & 3 & 1 & 5 \\ 1 & 0 & 2 & 1 \\ 3 & 4 & 1 & 8 \end{bmatrix}. \quad (1)$$

(Hint: You can stop performing row operations when you get an upper triangular matrix. Why?)

Solution.

$$\begin{bmatrix} 1 & 1 & 0 & 2 \\ 2 & 3 & 1 & 5 \\ 1 & 0 & 2 & 1 \\ 3 & 4 & 1 & 8 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - 2R_1}$$

$$\begin{bmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 2 & 1 \\ 3 & 4 & 1 & 8 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_1}$$

$$\begin{bmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & -1 & 2 & -1 \\ 3 & 4 & 1 & 8 \end{bmatrix} \xrightarrow{R_4 \rightarrow R_4 - 3R_1}$$

$$\begin{bmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & -1 & 2 & -1 \\ 0 & 1 & 1 & 2 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 + R_2}$$

$$\begin{bmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 3 & 0 \\ 0 & 1 & 1 & 2 \end{bmatrix} \xrightarrow{R_4 \rightarrow R_4 - R_2}$$

$$\begin{bmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Question 2. Suppose that

$$\varphi: \mathbb{R}^n \times \cdots \times \mathbb{R}^n \rightarrow \mathbb{R} \quad (2)$$

is an n -linear, alternating function. Show that

$$\varphi(\mathbf{v}_1, \dots, \mathbf{v}_n) = \varphi(\mathbf{v}_1 - \mathbf{v}_n, \mathbf{v}_2 - \mathbf{v}_n, \dots, \mathbf{v}_{n-1} - \mathbf{v}_n, \mathbf{v}_n). \quad (3)$$

for any vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n \in \mathbb{R}^n$.

Solution. By multilinearity we have

$$\begin{aligned} \varphi(\mathbf{v}_1 - \mathbf{v}_n, \dots, \mathbf{v}_{n-1} - \mathbf{v}_n, \mathbf{v}_n) \\ = \varphi(\mathbf{v}_1, \mathbf{v}_2 - \mathbf{v}_n, \dots, \mathbf{v}_{n-1} - \mathbf{v}_n, \mathbf{v}_n) - \varphi(\mathbf{v}_n, \mathbf{v}_2 - \mathbf{v}_n, \dots, \mathbf{v}_{n-1} - \mathbf{v}_n, \mathbf{v}_n). \end{aligned} \quad (4)$$

Focus on the second term on the right hand side. Notice that $\mathbf{v}_n$ appears in the first and last component. Therefore, by the alternating property this second term is zero. Continuing in this way, e.g. at the next step:

$$\varphi(\mathbf{v}_1 - \mathbf{v}_n, \mathbf{v}_2 - \mathbf{v}_n, \dots, \mathbf{v}_{n-1} - \mathbf{v}_n, \mathbf{v}_n) = \varphi(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 - \mathbf{v}_n, \dots, \mathbf{v}_{n-1} - \mathbf{v}_n, \mathbf{v}_n) \quad (5)$$

we eventually obtain the desired identity.